Project Report

ShareLocal : A Platform for Community Item Exchange

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Project Description

ShareLocal is a Django-based web application that connects people in local communities to share items they no longer need or want to lend. Whether you're giving away clothes, lending tools, selling gently used items, or renting out equipment, ShareLocal makes it easy to find people nearby who might need what you have to offer.

The platform focuses on building sustainable communities by reducing waste, encouraging reuse, and fostering neighborly connections through location-based sharing.

Software Requirements Specification

for

COMMUNITY ITEM EXCHANGE

SHARELOCAL

Prepared by

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Date: 23/03/2026

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Mahi Kubavat Heer Maheshwari Jay Makwana	Initial SRS prepared for SEPP course project.	23/03/26

1 Introduction

1.1 Document Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of **ShareLocal**, a **Community Item Exchange**. It serves as a formal reference for understanding the system's objectives, features, and constraints and is intended for use by developers, testers, instructors, and other stakeholders involved in the project. This document covers the complete scope of the system and provides a clear basis for system design, implementation, and evaluation.

1.2 Product Scope

ShareLocal is a web-based **Community Item Exchange** platform developed so that it enables users to share, sell, or borrow unused items within their local community. The platform aims to reduce wastage, promote sustainability, and provide cost-effective access to goods.

1.3 Intended Audience and Document Overview

This document is intended for:

- Faculty and Project Evaluators
- Software Developers
- Testers
- QA Engineers
- Documentation Writers

The document is organized into sections covering system overview, functional requirements by user roles, non-functional requirements, system constraints, and assumptions. Readers are encouraged to begin with the Introduction and Overall Description sections before reviewing the detailed requirements.

1.4 Definitions, Acronyms and Abbreviations

Term / Acronym	Description
CIE	Community Item Exchange
ShareLocal	Name of the platform for Community Item Exchange
SRS	Software Requirements Specification
Admin	User responsible for managing system settings and users
UI	User Interface
DB	Database

1.5 Document Conventions

This Document was created based on the IEEE template for System Requirement Specification Documents.

1.6 References and Acknowledgments

Python Documentation:
https://docs.python.org/3.14/tutorial/
Djano Documentation:
https://docs.djangoproject.com/en/6.0/intro/tutorial01/
MySQL Documentation:
https://dev.mysql.com/doc/

2.Overall Description

2.1 Product Overview

ShareLocal is a community-based web application developed to facilitate the exchange of unused or rarely used items among people living in nearby locations. The system is designed as an online platform that allows users to list items for sharing or selling and enables other users to search and request those items based on their needs.

The platform aims to reduce unnecessary purchasing, minimize household clutter, and promote sustainable use of resources by encouraging reuse within local communities. ShareLocal replaces informal and inefficient methods of item exchange with a structured, secure, and user-friendly digital solution. The system will be accessed through a web browser and will use a centralized database to store user profiles, item details, and request information.

2.2 Product Functionality

The **ShareLocal** system provides the following core functionalities to support community-based item exchange:

- **User Registration and Authentication**
  The system allows users to register and log in securely using valid credentials. Authenticated users can access system features according to their role.

- **User Profile Management**
  Users can view and update their personal profile information, including contact details and location, to enable local item discovery.

- **Item Upload and Listing Management**
  Item sellers can upload unused items by providing details such as item name, category, description, price (if applicable), and availability type (lend or sell). Sellers can edit or remove their listings at any time.

- **Item Search and Filtering**
  Item buyers can search for items using filters such as category, price range, and location to find suitable items within their local area.

- **Item Request Handling**
  Buyers can send requests for items they are interested in. Sellers can view incoming requests and accept or decline them based on availability.

- **User Communication**
  The system enables buyers to contact item owners through the platform to discuss item details and exchange information securely.

- **Category Management**
  Items are organized into predefined categories to ensure easy navigation and efficient searching.

- **Administrative Control**
  The system administrator can manage users and categories, monitor system activities, and maintain platform integrity.

- **Data Security and Management**
  The system ensures secure storage, access, and management of user and item data to maintain confidentiality and reliability.

2.3 Design and Implementation Constraints

The ShareLocal system will be developed as a web-based application using standard web technologies like Django and a relational database for data storage.The system shall support role-based access control to ensure secure operation. Development is constrained by academic timelines, limited development resources, and deployment on commonly available hardware and web browsers. The system must adhere to basic data security and privacy practices to protect user information.

2.4 Assumptions and Dependencies

The development of ShareLocal is based on the assumption that users have access to the internet and basic knowledge of using web applications. The system assumes availability of standard web browsers for accessing platform features.

The platform depends on the proper functioning of the web server and database system for storing and retrieving data. Any changes in project requirements, technology stack, or hosting environment may impact system design and implementation.

3.Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

ShareLocal provides a **web-based graphical user interface** that allows users to interact with the system through standard web browsers. The interface is designed to be simple, responsive, and user-friendly so that users with basic technical knowledge can easily use the platform.

The major user interface screens include:

- Home page with platform overview and navigation
- User registration and login pages
- User dashboard for managing profile and activities
- Item upload and item listing pages
- Item search and filtering interface
- Administrator dashboard for managing users and categories

All interfaces use clearly labeled buttons, forms, and navigation menus to reduce user errors and improve usability.

3.1.2 Hardware Interfaces

ShareLocal does not require any specialized hardware. The system can be accessed using standard computing devices such as desktops, laptops, tablets, or smartphones equipped with an internet connection and a web browser.The application is hosted on a server that stores system data and processes user requests.

3.1.3 Software Interfaces

ShareLocal The system interacts with:

- A **web browser** to provide user access

- A **MySQL Database system** to store user profiles, item details, and request information.
- A **GPS API** to search nearby items.

All software interactions are handled through secure and well-defined interfaces to ensure reliable data exchange and system stability. We use python and Django for developing the project.

3.2 Functional Requirements

R.1:Manage User

Description:The user registration function allows a new user to create an account on the ShareLocal platform by providing valid personal and login details. The system verifies the information and creates a new user account.The login function authenticates users by verifying and allowing access to the system if valid.

R.1.1: Enter Registration Details

Input:User enters name, email address, password, and location
Processing:System validates the input data and checks for existing email
Output:Users are registered into the system.

R.1.2: Select Login Option

Input: User selects the “Login” option
Output: System prompts the user to enter email and password

R.1.3: Authenticate User

Input: User enters email and password
Processing:System verifies credentials against stored data
Output: Successful login or error message

R.2: User Profile & Dashboard Management

Description: The dashboard function allows users to maintain their personal information, track their activity history, and manage the items they have listed on the platform.

R.2.1: View Profile

Input: User selects the “Profile” option
Processing: System retrieves user profile information
Output: User profile details displayed (name, email, location, etc.)

R.2.2: Edit Profile

Input: User selects “Edit Profile” and updates name, location, or profile picture.
Processing: System validates the new data and updates the user profile in the database.

Output: Profile successfully updated message.

R.2.3: View Transaction History

Input: User selects “Transaction History.”
Processing: System retrieves all past completed requests (borrowed, lent, bought, or sold).
Output: A chronological list of previous transactions with status details.

R.3: Manage Items

Description:The Manage item function allows users to list unused items for lending, selling, or free sharing.:It enables users to find items available in their local area.This function updates the availability status of an item after transaction completion.

R.3.1 Upload Item

R.3.1.1: Select Upload Item Option

Input: User selects “Upload Item” option
Output:System prompts user to enter item details

R.3.1.2: Enter Item Details

Input: User enters item name, description, category, condition, location, and item type
Processing: System validates and stores item information
Output:Item listing successfully created

R.3.2: Search Item

R.3.2.1: Select Search Option

Input: User selects “Search Item” option
Output: System prompts user to enter search keywords

R.3.2.2: Enter Search Criteria

Input:User enters keyword and filter options
Processing:System searches database and filters results
Output:List of matching items displayed

R.3.3 View Item

R.3.3.1: View Uploaded Items

Input: User selects “My Listings.”
Processing: System fetches all items uploaded by the user.
Output: List of items displayed with options to modify or remove.

R.3.4: Mark Item as Unavailable

R.3.4.1: Select Update Status Option

Input:Owner selects “Mark as Unavailable”
Processing: System updates item status
Output:Item removed from search results

R.4: Manage Item Request

Description: The send request function allows users to request an item for borrowing or purchasing.The manage requests function allows item owners to accept or reject incoming requests.

R.4.1: Select Request Option

Input: User selects “Request Item” option
Output:System prompts for confirmation

R.4.2: Submit Request

Input: User confirms request
Processing: System records request and notifies item owner
Output: Request submission confirmation

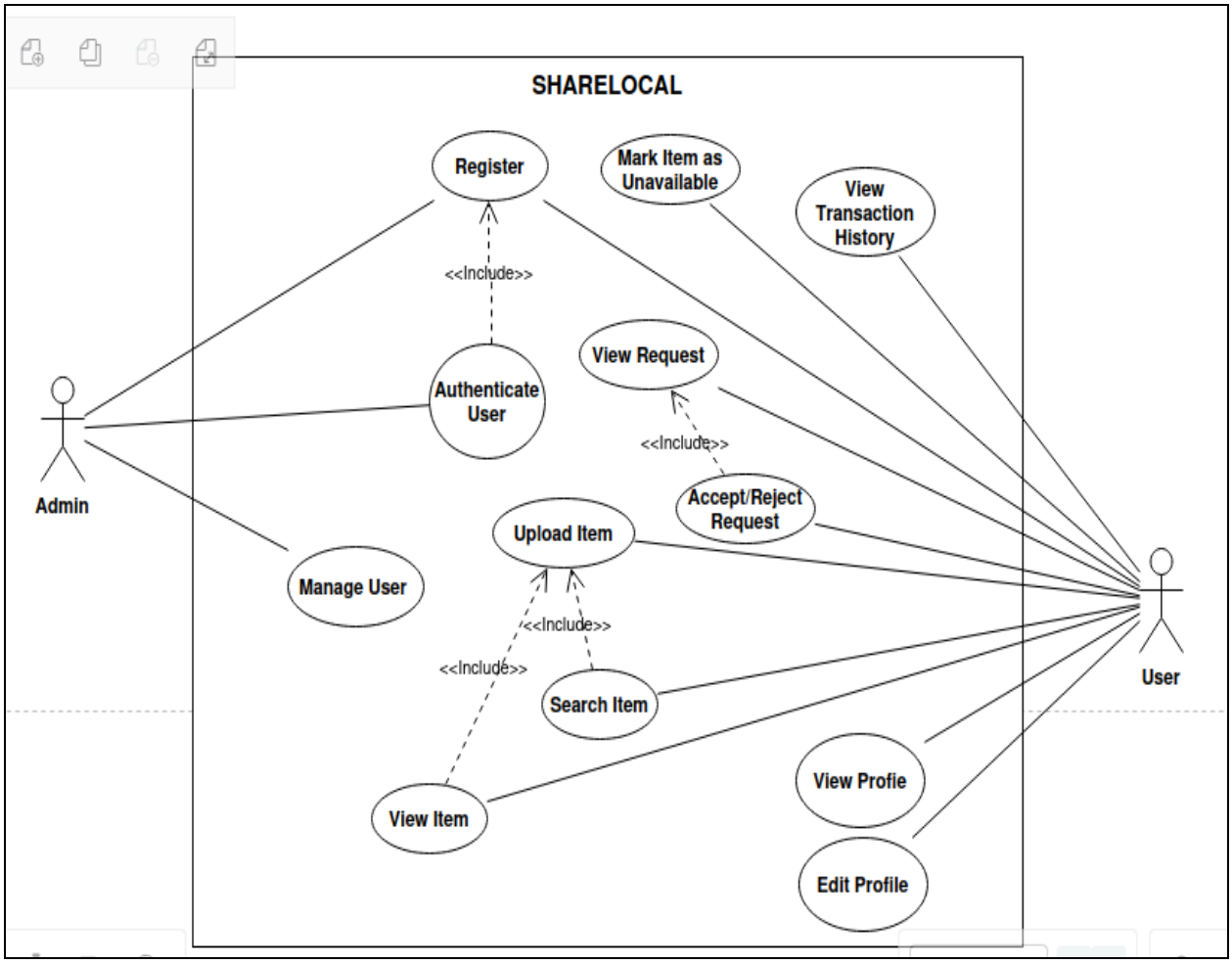
R.4.3: View Requests

Input:Owner selects “View Requests” option
Output:System displays pending requests

R.4.4: Accept or Reject Request

Input: Owner selects accept or reject
Processing: System updates request status
Output:Request status updated and users notified

3.3 Use Case Model



4.Other Non-functional Requirements

4.1 Performance Requirements

The ShareLocal system shall provide acceptable response times for all user operations such as login, item search, item upload, and request processing under normal operating conditions. The system shall support multiple users accessing the platform simultaneously without significant performance degradation.

4.2 Safety and Security Requirements

The ShareLocal system shall ensure secure access through user authentication and authorization mechanisms. Only registered and authenticated users shall be allowed to access platform functionalities such as item upload, requests, and communication.

4.3 Software Quality Attributes

4.3.1 Reliability

Core functionalities such as item listing, searching, requesting, and profile management shall execute correctly without data inconsistency or loss.

4.3.2 Usability

The ShareLocal system shall provide a simple, intuitive, and user-friendly interface suitable for users with basic technical knowledge.

4.3.3 Maintainability

The system shall be designed using a modular architecture so that updates or changes to one module (such as item management or user management) do not affect other system components.

4.3.4 Portability

The ShareLocal system shall be platform-independent and accessible through standard web browsers

4.4 DataBase Management

The ShareLocal system manages users and items into the database with consistency of data.

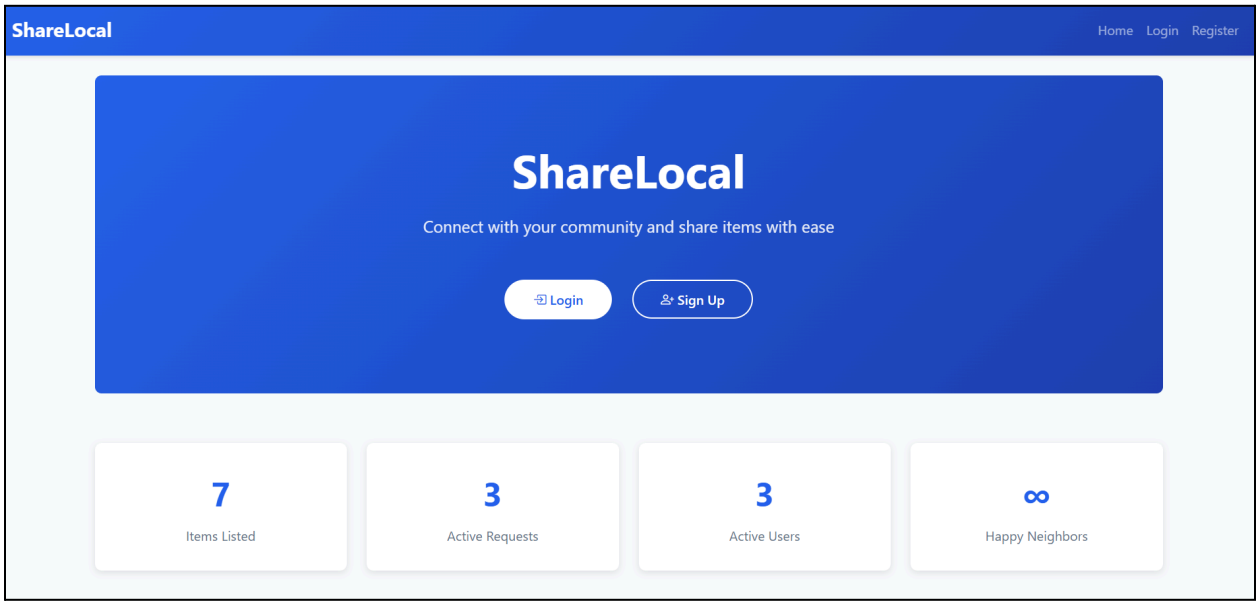
Implementation

of

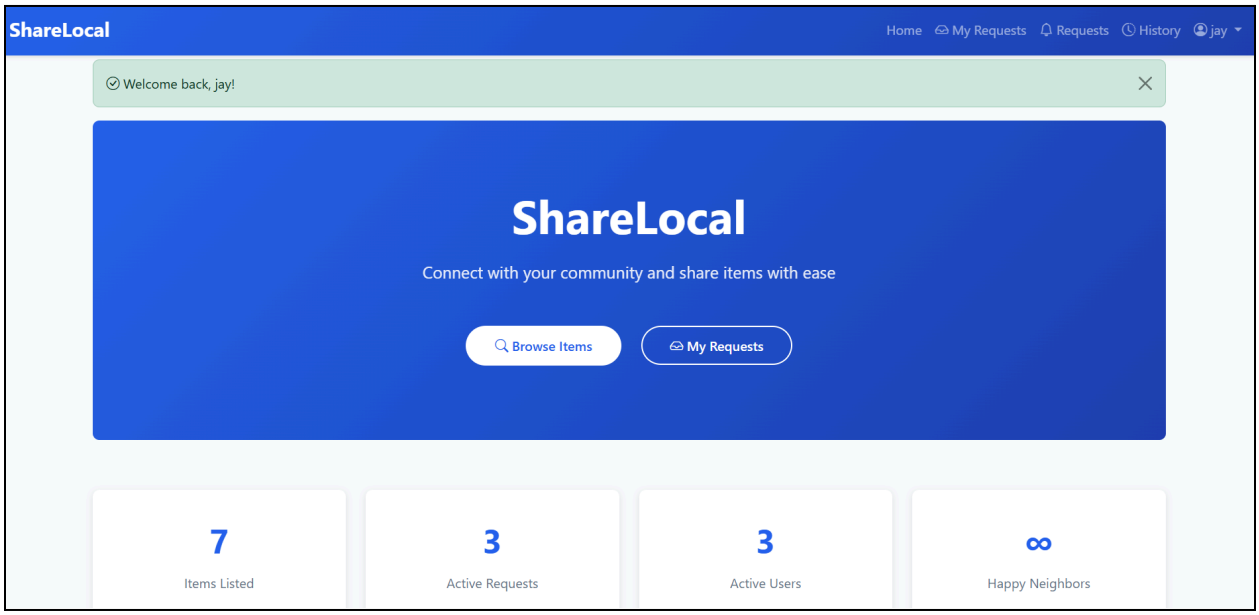
COMMUNITY ITEM EXCHANGE

SHARELOCAL

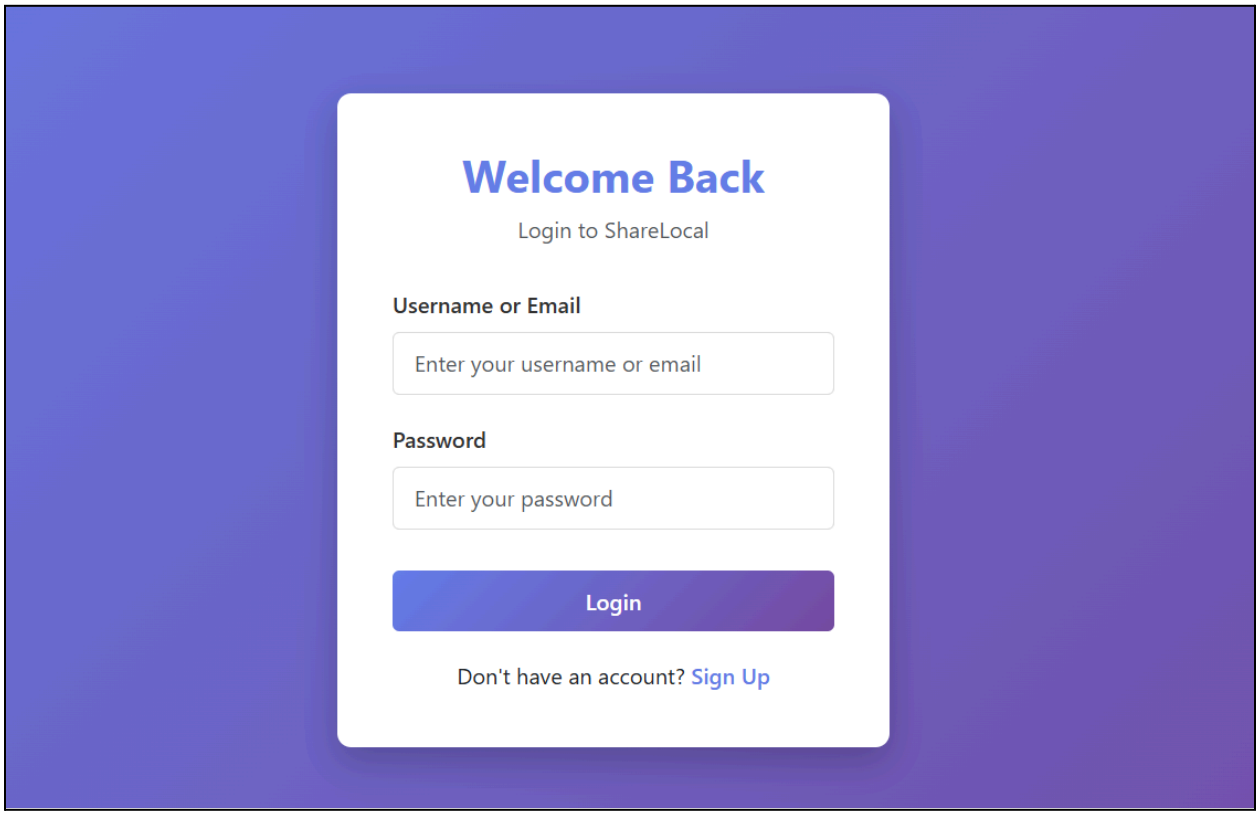
- The Home Page



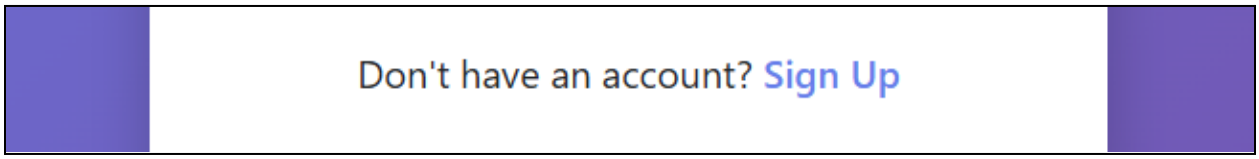
This is the home page of the system. If the user is not logged in, the user gets the option to login and if the user is already logged in, the user gets the home page as shown below.



- Login

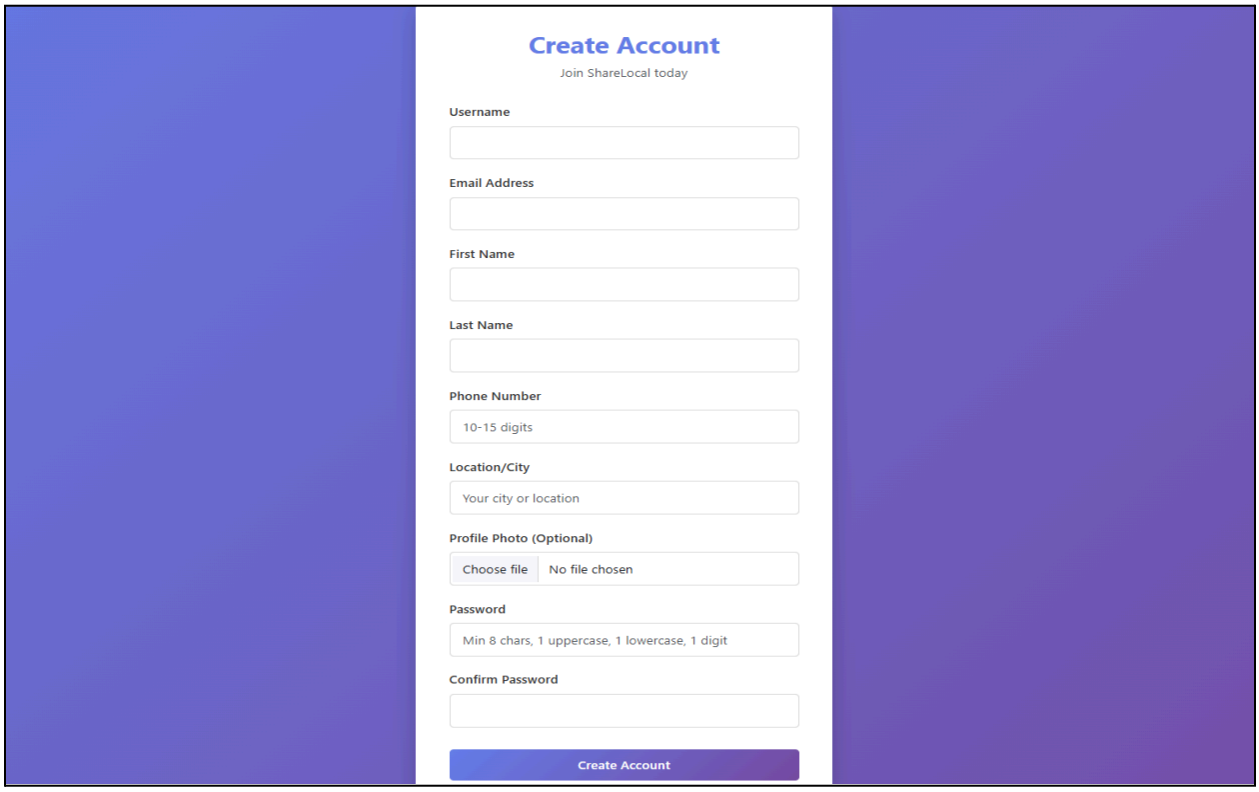


On the login page, the user can login in his/her account by inserting username & password.



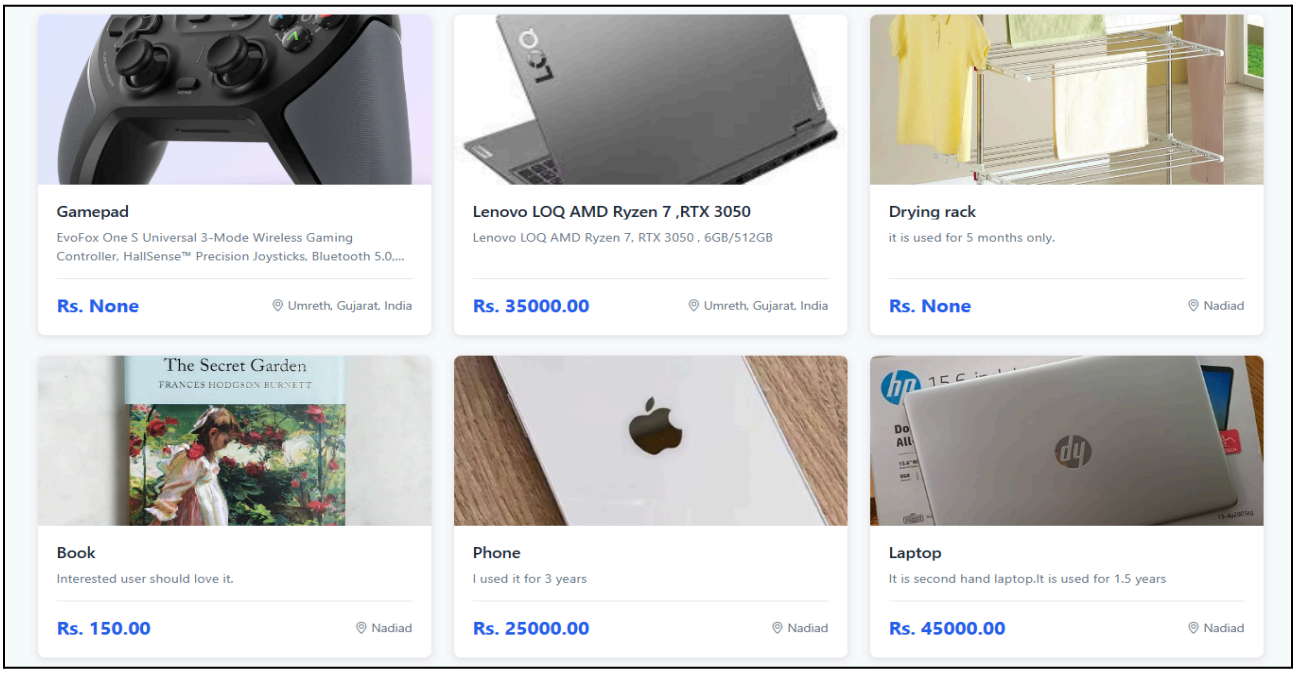
There is an option for registration(sign up) for users, who have not yet registered.

- Registration



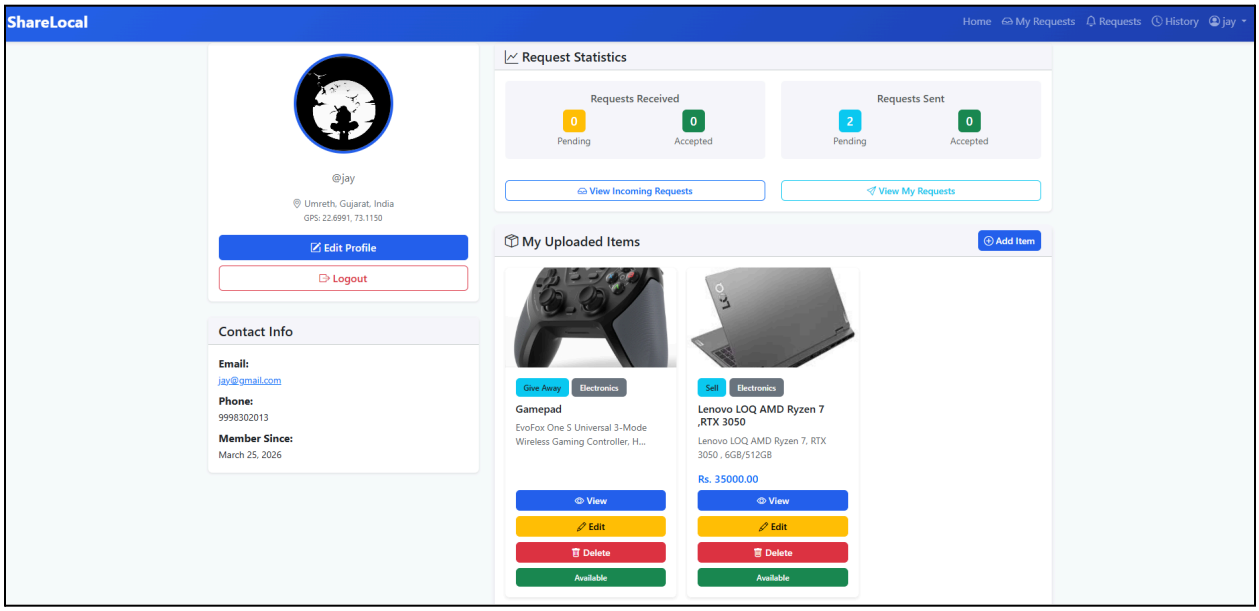
A user can register here using his/her details. And make the passwords according to constraints.

- Featured Items



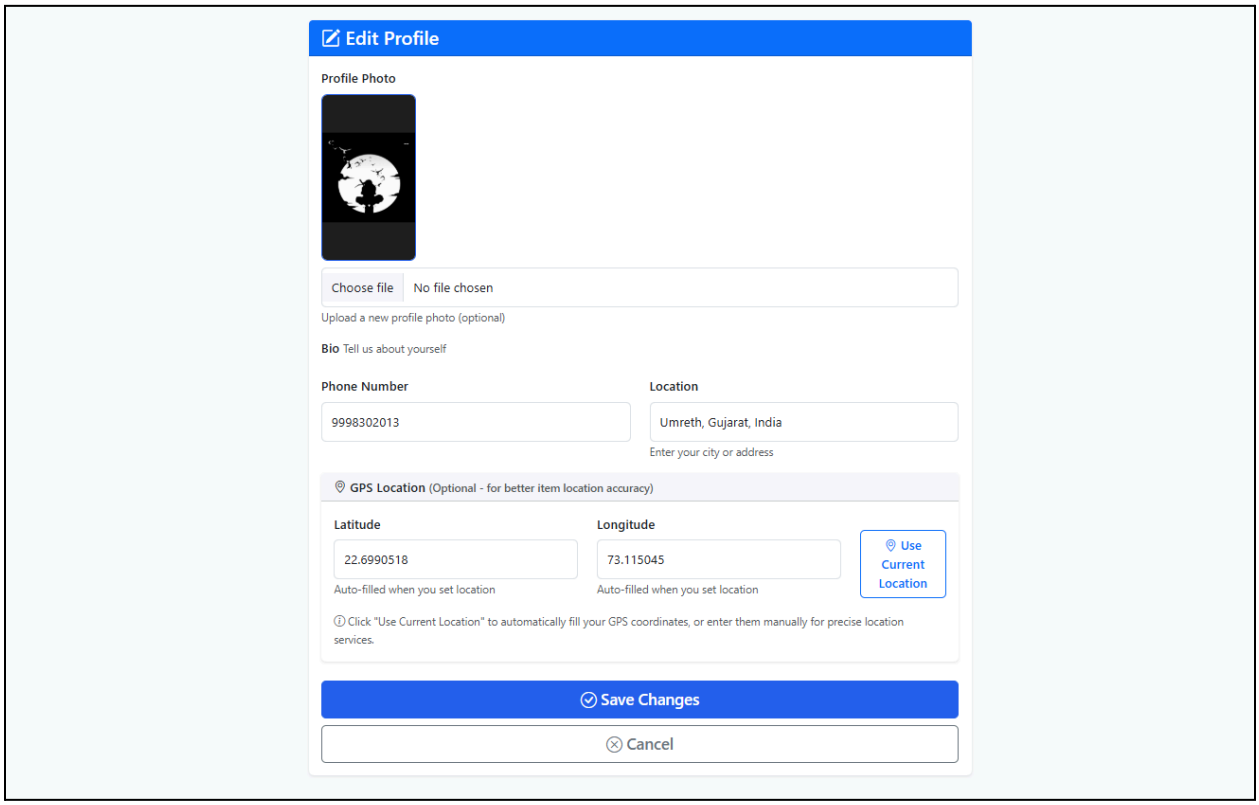
This shows 6 items recently uploaded.

- Profile



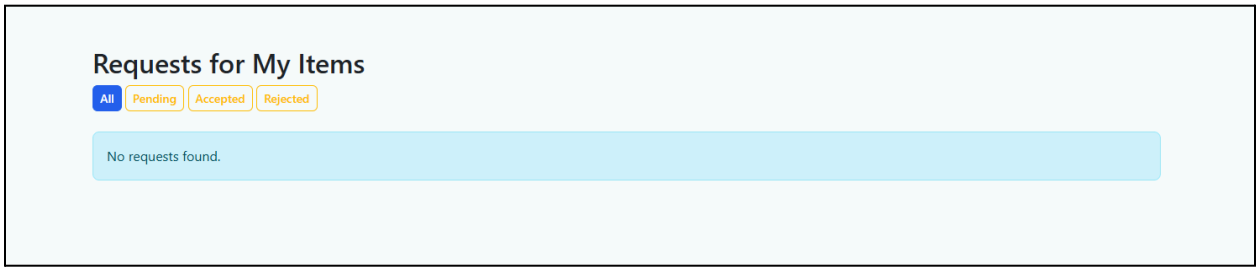
When a user is logged in he/she has a profile option. Here users can see the items uploaded by them. Requests sent for other's items and requests for their own items, also how many requests are pending and accepted .It also displays user's contact details and option to edit profile.

[Edit Profile]



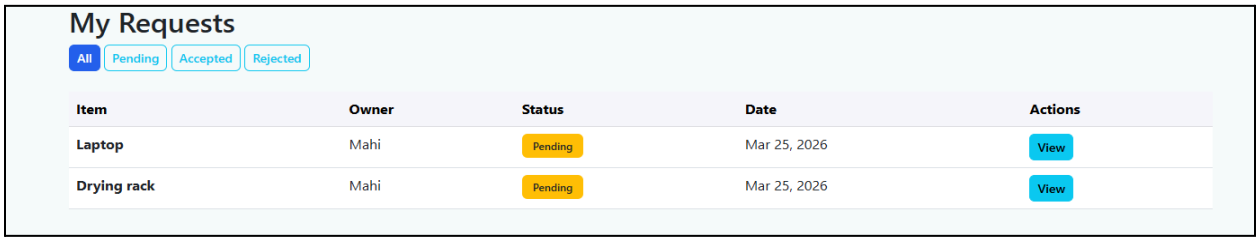
When the user clicks on the Edit button, he/she can see this page where the above details can be changed and saved in the system.

[Incoming Request]



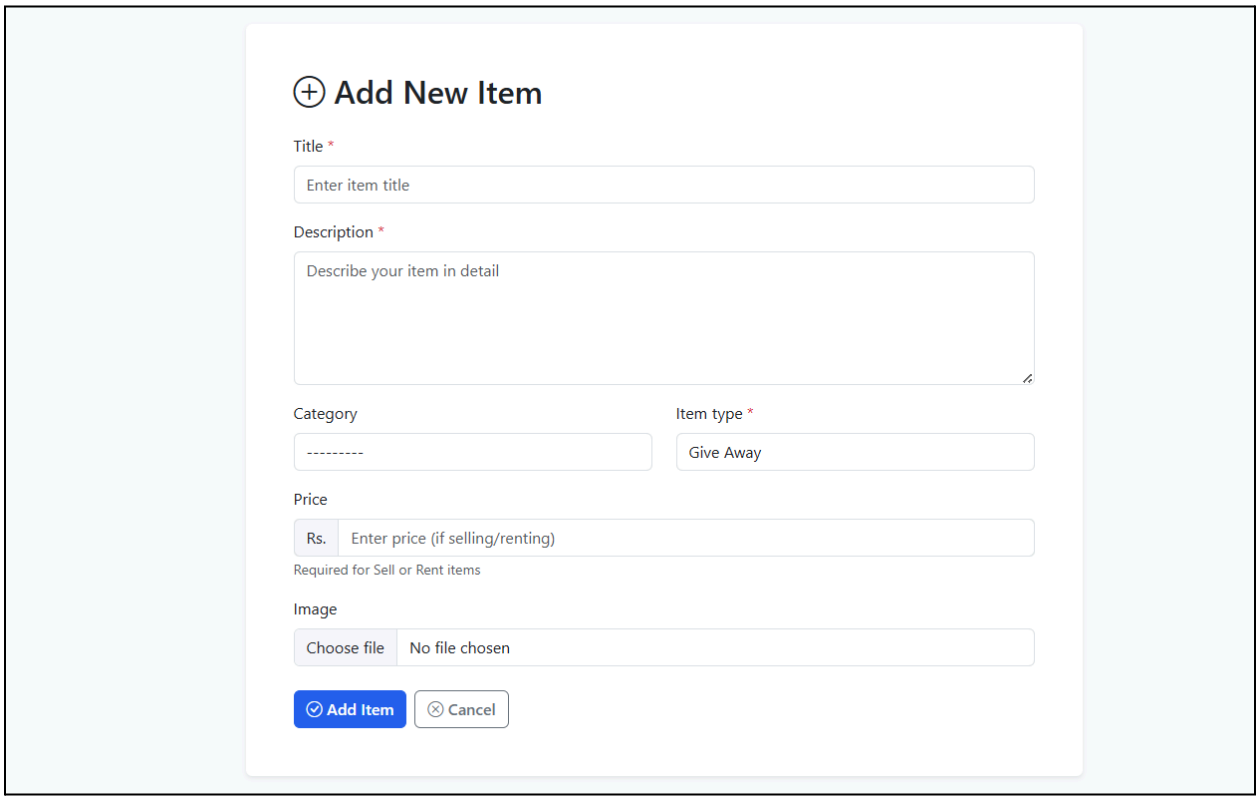
When the user clicks on the Incoming request button, he/she can see the incoming request and also the pending,accepted and rejected request.

[My Request]



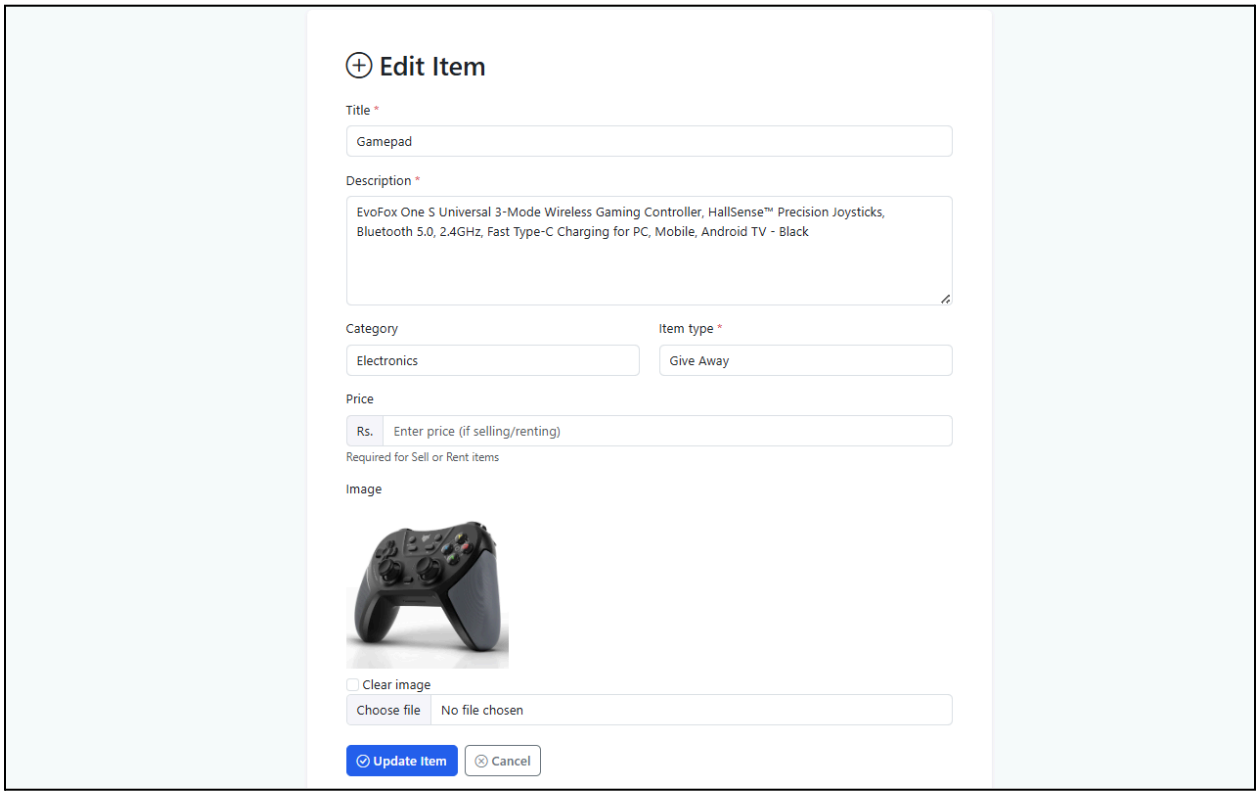
When a user clicks on My request, he/she can see the requests sent by him/her for other user's items.

- Add Item



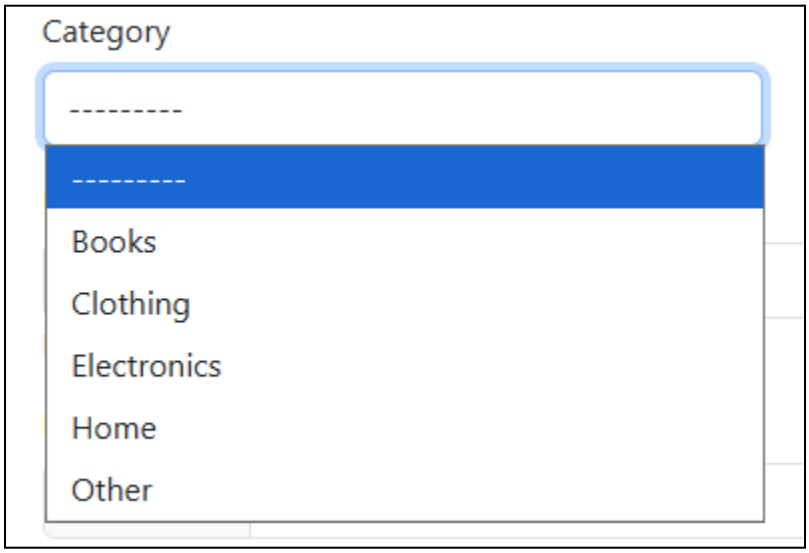
When the user wants to upload an item , the user is provided with a form which the user has to fill-up.

[Update Item]



Users can update previously uploaded items.

[Category]



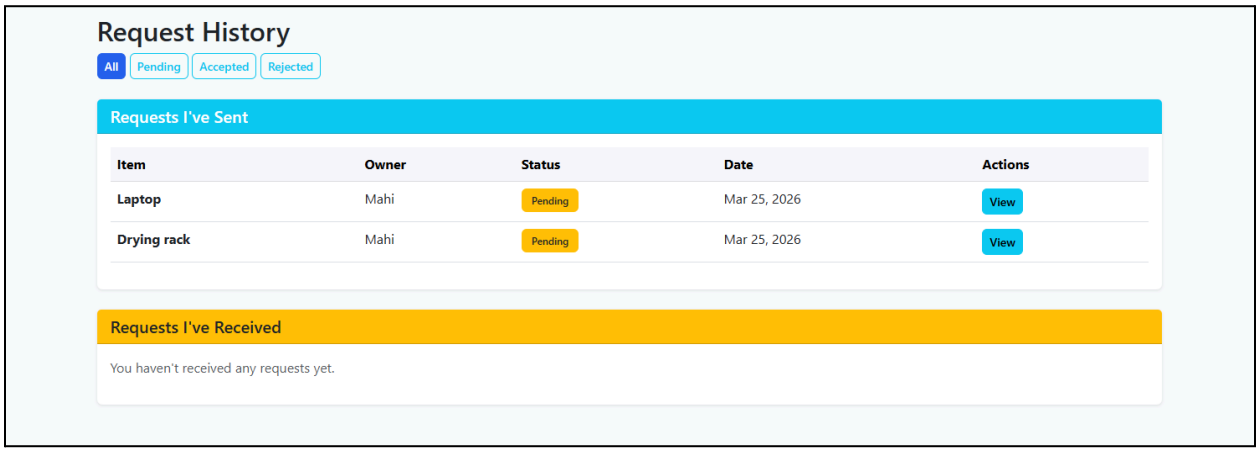
Users are provided with a category drop down menu, from which items's category has to be chosen for making items easy to search for other users.

[Item Type]

All the items are divided into 3 types :

1) Give Away : if the user wants to give item for free (by default all items are in Give Away Type)
2) Sell : If the user wants to sell the item.
3) Rent : If the user wants to give an item on rent.

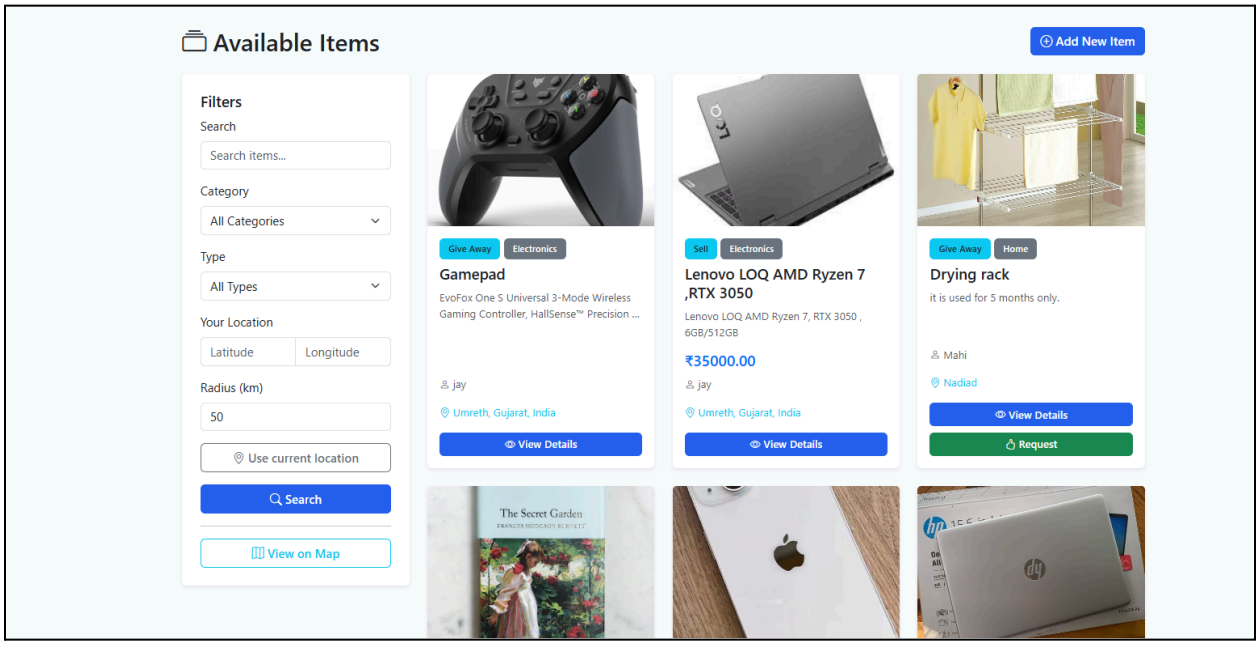
[Request History]



Shows all the requests sent and received.

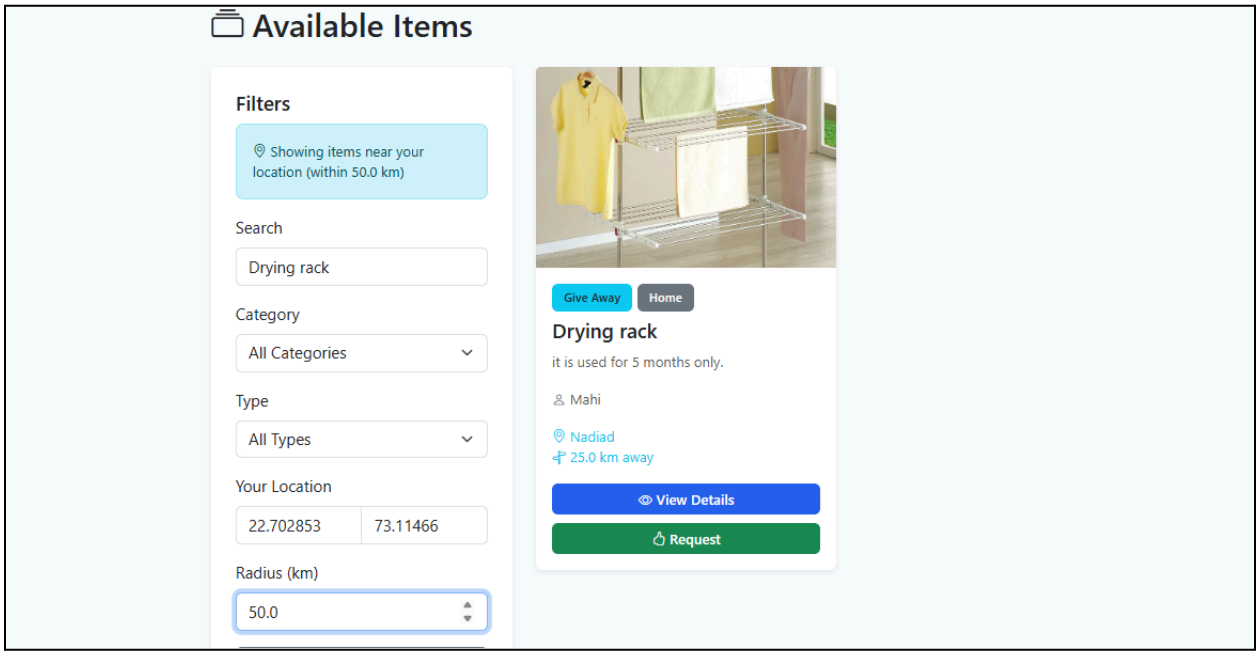
- # Browse Items

Here By default based on your location , items in the radius of 50km are shown.



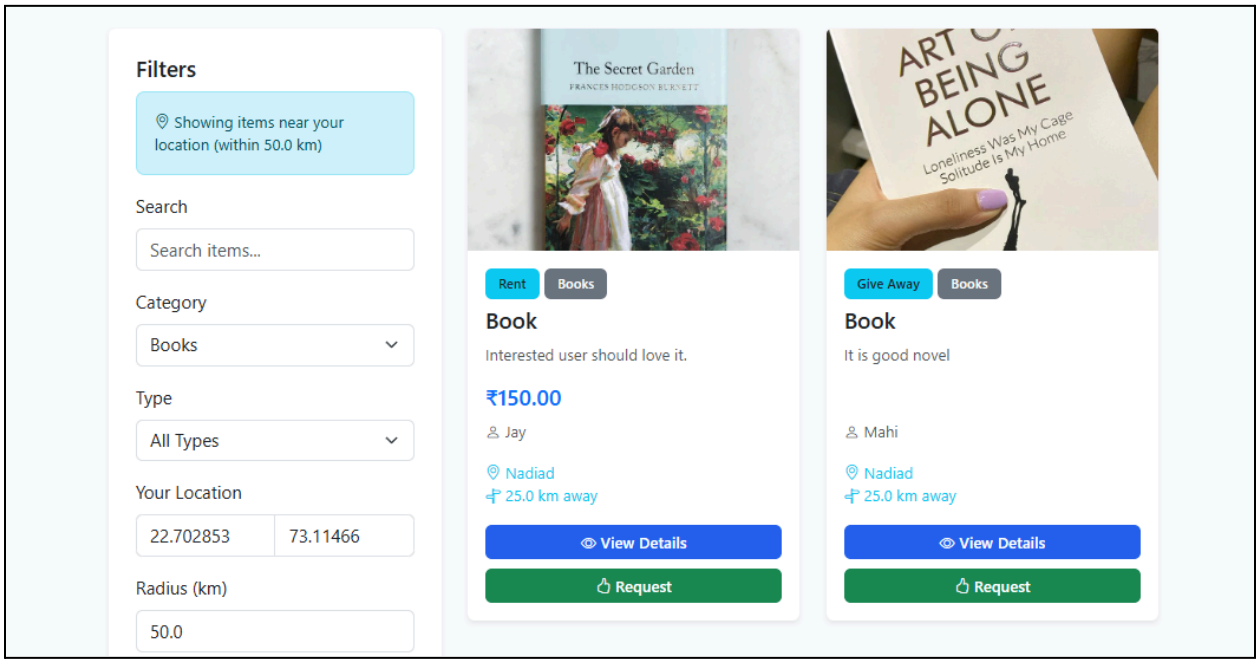
When the user clicks on the Browse Items ,the user can see all the items uploaded into the system and can request for items as well as search items using filters option.

[Search Filter]



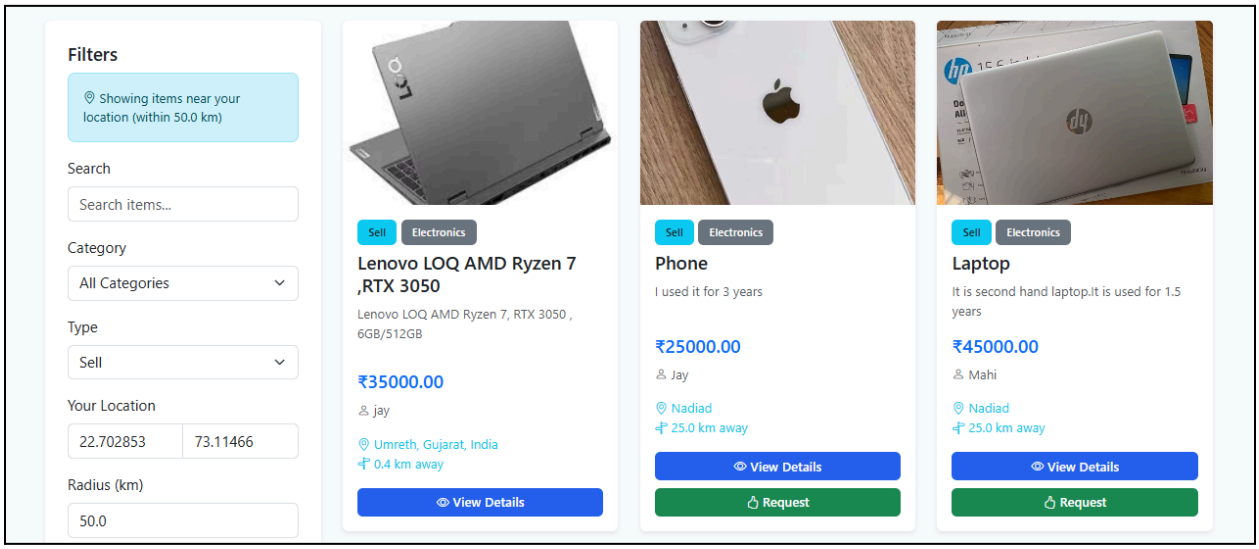
Users can filter items using the name of the item.

[Category Filter]



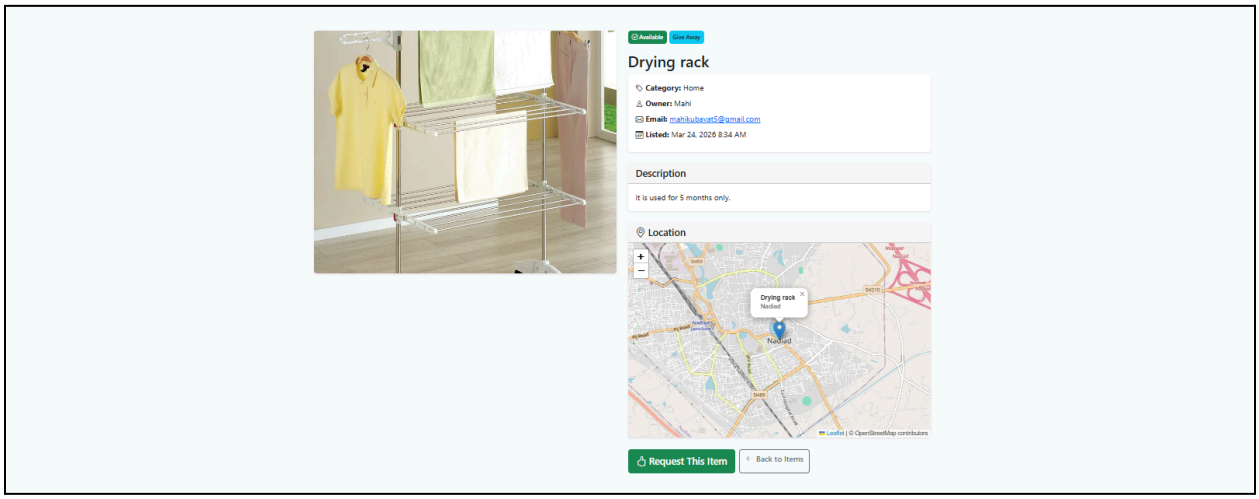
Users can filter the items based on their categories.

[Type Filter]



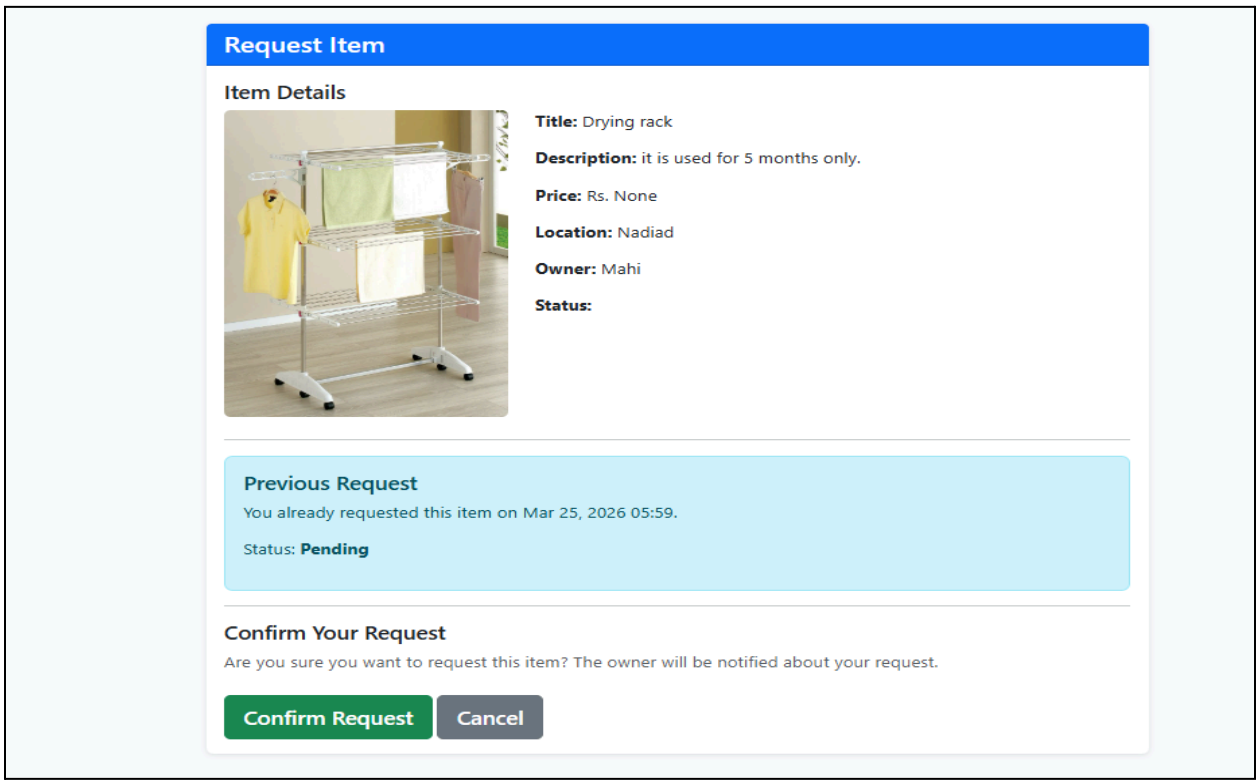
Users can filter the items based on their types (Give away/sell/rent).

[View Item]



When the user clicks on View Details , the user can see the item's details, owner's name and email as well as the item's location on the map.

[Request Item]



When clicked on a request item , the system asks for confirmation of the request. If an item was already requested previously it will show that the item already requested with time and status.

Software Project Lab Work

Django Poll App

Project Summary

Through the development of this application, we have successfully implemented a full-stack web project using the Django framework. By following the Model-View-Template (MVT) architecture, we have decoupled the data logic from the presentation layer, allowing for a cleaner and more scalable codebase. We have also mastered the use of Django’s Object-Relational Mapper (ORM), which enabled us to define our database schema using Python classes and perform complex data queries without writing manual SQL.

Furthermore, we have integrated a secure user-facing interface with a robust administrative backend. By utilizing Django's built-in security features, we have ensured that our application is protected against common web vulnerabilities like Cross-Site Request Forgery (CSRF). Finally, we have optimized the development workflow by implementing automated tests and utilizing generic views, which significantly reduced the amount of boilerplate code required to deliver a fully functional polling system.

Key Functionalities

Functionality	Short Explanation
Object-Relational Mapper (ORM)	Allows developers to define database tables as Python classes. It enables querying and updating data using Python code instead of raw SQL.
Database Migrations	A version control system for the database. It tracks changes to data models and propagates those changes into the database schema automatically.
Django Admin Interface	A pre-built, production-ready interface that reads your models to provide a visual backend for adding, editing, and deleting data.
URL Dispatcher	A clean "table of contents" system (urls.py) that maps browser URLs to specific Python functions or views.
Template Tags & Filters	Specialized syntax used within HTML files to inject dynamic data, perform loops (e.g., listing all polls), or logic checks (e.g., showing an error if a vote is invalid).
Generic Views	Abstracted classes that handle common web development tasks (like displaying a list or a detail page) with minimal code.
CSRF Protection	A built-in security feature that protects forms from "Cross-Site Request Forgery" attacks by requiring a unique token with every POST request.
Automated Testing	The use of Django's TestCase class to simulate user behavior and verify that logic—such as "polls should not appear until their publish date"—is working correctly.

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THANK YOU